Manuscript outline, checked against the draft of 2026-10-11

enceladus_DSM `studies/keff_sintering/manuscript_outline.tex`

2026-10-11

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,^{\circ}\mathrm{C}$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. Read against `Literature/BaglinoJackson_DSM_manuscript.pdf` (27 pages, 657 numbered lines). Paragraphs are identified by the draft's line numbers (L first–last). **[in draft]** covered; **[missing]** not there; **[revise]** there but no longer matches the model or results; **[incorrect]** a statement that is wrong as written; **[cut]** can go, or shrink, without loss. Boxes mark where a figure or table goes. Themes are prompts, not text.

Where the draft stands

Section	State
Key points	one placeholder
Abstract, plain-language summary	marked "needs to be updated"; both still claim "without retuning" and a collapse across porosity
1 Introduction	complete; kept as drafted. Three paragraphs repeat each other and one sets up a model the paper no longer has
2.1 Phase-field model	nearly complete; kinetic parameters undefined
2.2 Gibbs–Thomson	Eq. (7) now correct; still two equations and no clock
2.3 Homogenization	complete; the derivation can be halved; SSA is never defined
Numerical method; set-up of the aggregates	**no section**
3.1 Two-grain sintering	Molaro part written, no numbers; saturated-pair paragraph still a placeholder
3.2 Aggregates	Figure 7 text is new and much closer; the quantitative statements are still absent
4 Discussion, 5 Conclusion	unchanged: written for the earlier model

Fixed since the last check: Eq. (7) has $1/\alpha_c$; $T_0 = 273.15$ K; Table 1 carries $\Delta\beta$ and the right τ_{sub}; Figure 2 caption names the 0.05 and 0.95 contours; Figure 6 caption normalizes at $\theta = 30$; Figure 7 has a full caption; the 0.3% statement is in the text.

Most in need of work, in order: (1) Discussion and Conclusion; (2) the numbers in Section 3; (3) Section 2.2 and the two missing Methods subsections; (4) the saturated-pair paragraph.

Where to cut

The first half (to the end of Section 2) is 235 numbered lines of text plus 17 numbered equations. About 45 lines and four equations can go with no loss of content. The same three things are each said three times:

Said three times	Where
What the paper does, in order	end of Introduction (L166–179); Results opening (L292–307); Discussion opening (L400–410). Keep the first.
Wilson–Frenkel, α_c, $\sigma_0(T)$, Libbrecht	Introduction (L116–133); Discussion (L441–452 and L453–468). Keep one short version.
Bouvet and Braun tune α_c	Introduction twice (L149–153, L161–165); Discussion at length (L469–491). Keep the long one in the Discussion.

Cut	Lines now	Lines after
Introduction, kinetics paragraph (L116–133)	18	6
Introduction, repeated sentences (L62–65, L98–100, L134–137, L161–165)	14	3
Methods opening (L181–188)	8	3
2.1 first paragraph, what a phase field is (L190–197)	8	2
2.3 weak-form derivation, Eqs. (12)–(14) (L260–275)	16 + 4 eqs	4
Results opening (L292–307)	16	3
3.2 homogenization restated (L363–367)	5	1
Discussion opening (L400–410)	11	0
4.1 model history (L412–424)	13	0

Wrong or inconsistent throughout

- **Notation.** K_{eff} in the text, k_{eff} in Figures 6–7, the tables and Section 3.2 from L383; K_{ice} against k_{ice}; ρ_i against ρ_{ice}; ρ_a, P_a (Eq. 2) against $\rho_{\text{air}}, P_{\text{atm}}$ (Table 1); ϕ_a called "water vapor" (L200), it is the air phase; ϕ, ϕ_i and $\boldsymbol{\phi}$ used interchangeably in Section 2.3.
- **Temperature.** The text has T in °C with $T + T_0$ (Eqs. 3, 5); Table 1 writes $D_{v0}(T/273.15)^{1.81}$ and $k_B T$, which need kelvin. Pick one.
- ε is called "the interface width" (L208); Table 2 calls it the interface parameter and gives the band as $9.2\,\varepsilon$. Use the table's wording.
- **Durations and counts.** "28-day" (L171), 30 everywhere else; mesh "2829×2928" (L349); "3 packings for each porosity" (L382); "25 microstructures" (L405) is right, "four microstructures" (L539) is not.
- **Broken references.** "??" at L264, L450, L462, L519, L527, L530, L538 and in the notes of Tables 1 and 2; about 20 citations print as "?" in Section 4; "table table 3", "section section 4.1", "figure figure 6".

- **Words.** "deposits on the neck" (L103): desublimates. "inhabit extraterrestrial life" (L84). "without less of generality" (L248): periodicity is an assumption, not a convenience. "Allen–Kahn" (L209). "tonsorial" (L289). "effect" for "affect" throughout.

Front matter

- **Title.** Kept; drop "Version 2".
- [missing] **Key points** (three, each $\leq$ 140 characters). (i) For a given packing, temperature only rescales time. (ii) Conductivity follows specific surface area by one power law in well-connected packings; porosity sets the level. (iii) Vapor transport alone gives part of the neck growth measured in the laboratory.
- [revise] **Abstract** (L9–31). Sentences 1–5 are background; keep two. Remove: "temperature gradients" as a driver (the runs are isothermal); "without retuning"; "temperature dependence ... measured in experiments"; collapse "for both varying temperatures and varying porosities". Order: setting (2 sentences); model and what it carries (1); pairs, with 41% and the exponent (2); aggregates, with the three numbers 0.27%, -0.80 and the porosity range (3); what it means (1).
- [revise] **Plain Language Summary** (L32–55). Remove the tuning story (L48–51) and "regardless of temperature or initial porosity".

1 Introduction

Kept as drafted (decision 2026-10-10); no reordering proposed. Paragraph by paragraph, with what could be trimmed in place:

1. **What dry snow metamorphism is** (L56–67). [in draft] [cut] The ice-cream, snow and avalanche examples (L62–65) return in the next paragraph; keep them in one.
2. **Why it matters on Earth** (L68–80). [in draft]
3. **Icy bodies and Enceladus** (L81–100). [in draft] [cut] The last sentence (L98–100) is circular: a better understanding of bulk properties aids a better understanding of bulk properties.

   Figure 1 [in draft] placed. [revise] The caption, and the copy of the figure in the draft, still say "neck of radius $R(t)$"; the label was changed to $w(t)$, and the Results use w as the neck *width*.

4. **Sintering as part of metamorphism** (L101–115). [in draft] "deposits" $\rightarrow$ desublimates. Demmenie is cited here; say in one clause that it is used later.
5. **Attachment kinetics** (L116–133). [revise] [cut] The longest paragraph of the Introduction builds toward a temperature-dependent α_c through $\sigma_0(T)$. The model holds α_c constant, so the reader is set up for something the paper does not do. Keep: the definition of supersaturation, the Wilson–Frenkel law, that α_c is uncertain over two decades. End on the choice made here: α_c held constant within each study. [incorrect] L125–127 says the barrier puts the kinetics "well below the vapor transport limited regime"; it means growth is limited by attachment, not by transport.

6. **Metamorphism beyond a pair; earlier models** (L134–153). [in draft] [cut] L134–137 restates paragraph 4. Bouvet is 2022 here and 2023 at L162.
7. **What is not known** (L154–165). [revise] [cut] L161–165 repeats L149–153. L160–161 claims no model predicts evolution "for varying temperatures and varying vapor saturation"; this paper does not close that gap, so do not open it. The gap the paper does address is L157–159: bulk properties without compaction, and the roles of temperature and porosity.
8. **What follows** (L166–179). [revise] "28-day"; "is dependent of"; the stray fragment L178–179; "validate model" (compare); the saturated pair is not announced; the promise of independence from "initial porosity" is wider than the result.

2 Methods

Opening (L181–188). [cut] The free-energy derivation and the coupling are restated in the first lines of 2.1. Keep the lineage (Kaempfer & Plapp; Moure & Fu) and the sentence that this is an application of the model.

2.1 Phase-field model for dry snow metamorphism

1. **What a phase field is** (L190–197). [cut] Textbook; one sentence.
2. **Unknowns** (L198–204). [in draft] ϕ_a is air, not water vapor.
3. **Scope.** (new) [missing] Of dry snow metamorphism the model carries sintering by vapor transport only. Not in it: surface diffusion, densification, overburden. Define "desublimation". Why the laboratory analogues are unloaded ice-sphere samples and not natural snow.
4. **Eq. (1) and the phase equation** (L205–211). [in draft] [missing] M_{sub} and α_{sub} are named but never given: state them in terms of τ_{sub} and ε, or point to Moure & Fu. [missing] The temporal scaling ξ_T, ξ_v is in Tables 2 and 3 and nowhere in the equations.
5. **Saturation vapor density, Eqs. (2)–(3)** (L212–216). [revise] Check Eq. (3) against the source: the last term reads $k_5 + \ln(T + T_0)$ and is probably $k_5 \ln(T + T_0)$. The text cites Flatau and Wexler; the note to Table 1 says ASHRAE.
6. **Energy equation and mixture rules** (L217–223). [in draft]
7. **Vapor equation and diffusivity** (L224–229). [in draft] [cut] T_0 is defined at L215 and again at L228.

> **Table 1** [in draft] Material properties, current version. [revise] Its note reads "Tables ?? and 3". It defines β_{sub}, $\Delta\beta$ and τ_{sub} one page before the text introduces them, and the text never mentions the last two.

2.2 Interface kinetics and the Gibbs–Thomson condition

1. **The condition and α_c** (L230–241). [in draft] Eqs. (6) and (7), now correct. d_0 is defined only in Table 1.
2. **The thin-interface limit.** (new) [missing] L234–235 says the model reproduces Eq. (6) "under the conditions outlined by Kaempfer & Plapp". Those conditions are where $\Delta\beta$ comes from: the realized coefficient is $\beta_{\mathrm{sub}} + \Delta\beta$. Two or three sentences.

3. **The clock.** (new) [missing] $\tau_{\text{sub}} = \varepsilon^2(\beta_{\text{sub}} + \Delta\beta)/d_0$ and the sintering age $\theta = t/\tau_{\text{sub}}$, defined here. At present θ first appears at L377 as "the characteristic relaxation time of the phase-field equation". State that ε is the same in every aggregate run. *Open: state the clock as τ_{sub} at fixed ε, or as $\tau_R = \bar{R}^2\beta_{\text{sub}}/d_0$.*

4. **The values used.** (new) [missing] $\alpha_c = 1.020 \times 10^{-3}$ (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair), held constant in temperature, each with its β_{sub}.

> **Table (temperature)** [missing] `\TableKeffTemperature`: β_{sub}, $\Delta\beta$, τ_{sub} and the final age at each temperature. Tables 1 and 2 both point to it ("Table ??"). It is also what explains why $-40\,^\circ$C is absent from Figure 7(b): that run ends at $\theta = 43$.

2.3 Homogenization for the effective thermal conductivity

1. **Two scales and the microscale problem** (L243–252). [in draft] "without less of generality".
2. **Macroscale law, $\mathbf{K}_{\text{eff}}$, cell problem** (L253–262). [in draft] [revise] The sentence at L257 breaks off ("referred to as co"). [incorrect] Eq. (11d) has $\mathbf{t}$ where it needs $\nabla\mathbf{t}$. [cut] L260–262 says the equations are coupled to their own boundary conditions.
3. **Weak form and the single-domain problem** (L263–275). [cut] Eqs. (12a)–(14) can become one sentence: summing the weak forms over the two subdomains, the interface terms cancel and Eq. (15) follows. Saves four equations.
4. **Diffuse form and the tensor law** (L276–290). [in draft] [missing] $\mathbf{K}_{\text{eff}}$ is defined for the sharp problem only (Eq. 10); give the diffuse form, the one actually evaluated.
5. **What is reported.** (new) [missing] The scalar $k_{\text{eff}} = (k_{xx} + k_{yy})/2$ is defined only in the caption of Figure 6. **SSA is never defined**: say how it is measured from ϕ and in what units, since half the results are plotted against it.

> **Figure 2** [in draft] placed. [revise] Not cited in the text. Caption says "red"; the lines are orange. Add that only the x problem is shown.

2.4 Numerical implementation

- [missing] The whole subsection, one paragraph: quadratic C^1 splines, time stepping and the step cap, sampling of $\mathbf{K}_{\text{eff}}$.
- [missing] One sentence each, with a pointer to the supplement: domain size (Fig. S1), interface parameter (Fig. S2), time step and sampling (Fig. S3). The second matters for the Results: the first day depends on ε.

> **Table 2** [in draft] The complete parameter table is in the main text, but [revise] it is cited nowhere. [cut] It takes most of a page. The shorter `\TableKeffParameters` holds what a reader needs; the complete one can go to the supplement.

2.5 Aggregates: packings, boundary conditions, simulation matrix

- [revise] Now inside Section 3.2 (L345–353). Move here.

- [in draft] Drop-and-roll deposition. [missing] Five packings per porosity, independent across porosity; the same packings at every temperature; grain size distribution.
- [missing] Why periodic boundaries: scale separation, no wall physics to invent, a closed system driven by curvature alone.
- [missing] Two dimensions, and why (now L303–304, in the Results opening).
- [missing] The simulation used for single-run figures: $\varphi = 0.325$, $-20\,^{\circ}\mathrm{C}$.

2.6 Two-grain set-up

- [revise] Now inside Section 3.1 (L315–327). The geometry, the axisymmetric formulation and the boundary conditions are Methods.

> **Table 3** [in draft] placed. [missing] No column for the saturated pair: its grain radius (84.4 μm), α_c, temperature and 100 h duration are stated nowhere. [revise] "Simulated time 2 h" at $-5\,^{\circ}\mathrm{C}$: the run with the thin-interface term reached 68 min, which covers the 48 min of data.

3 Results

Opening (L292–307). [cut] [revise] Third statement of the plan. If kept, three sentences. [incorrect] "independent of the mean temperature as well as the initial porosity" (L302–303). Demmenie is described as theory only; they measured the exponent too. The two-dimensional restriction belongs in the Methods.

3.1 Two-grain sintering

[revise] Title "Phase-Field Model Validation": the model gives 41% of the measured growth at one temperature and 170% at the other. "Comparison with two-grain experiments" claims no more than is shown.

Unsaturated pair (Molaro et al., 2019).

1. **Why this experiment** (L309–314). [in draft]
2. **Set-up** (L315–327). [revise] Move to the Methods; keep here only the humidity inference and the choice of α_c. [incorrect] "The only free parameter in our model is β_{sub}" (L322–323): the humidity is fitted too. Give $\alpha_c = 0.1$ as a number. "(cite other works as well)".
3. **Figure 3** (L328–338). [in draft] panels (a), (b). [missing] The numbers: 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$ over 78 min; 170% at $-5\,^{\circ}\mathrm{C}$ over 48 min (52.1 against 44.0 μm). [missing] Panels (c) and (d): at $-20\,^{\circ}\mathrm{C}$ the shrinkage is what the humidity was fitted to, so it is a calibration and not a test. [missing] What the colors in (a) show (σ).

> **Figure 3** [in draft] placed. [revise] Caption stops at (b).

Saturated pair (conditions of Demmenie et al., 2025).

1. **The paragraph** (new) (L339–340 is "Insert paragraph"). [missing] Why a second comparison: at saturation the grains do not shrink, so neck growth by vapor transport is isolated. Equal

grains of 84.4 μm at $-20\,^{\circ}$C for 100 h; grain radius changes by +0.06%; neck width 28 to 63 μm; exponent 0.21 after the relaxation (0.21–0.23 depending on the window), against 0.26–0.33 measured and 1/3 in theory.

2. *Open: how to read the low exponent. Two readings are on the table. (i) Yours: mechanisms absent from the model. (ii) Mine: the run is faster than the Gibbs–Thomson prediction early, where the neck fillet is about two interface bands across, and that flattens the fitted slope; the prediction for this run gives 0.29 and the final width to 4%. The paragraph can report the number without choosing.*

Figure 4 [in draft] placed. [revise] Not cited in the text. Caption gives no temperature, grain size or α_c, and no fitting window.

3.2 Aggregates

1. **Set-up and the gallery** (L342–360). [revise] Two paragraphs in one. The set-up (L345–353) goes to the Methods. [incorrect] "We initially consider a single 2D microstructure" (L350) after stating 125 runs. The gallery description (L354–360) is fine. [missing] What the colors show: where ice sublimates and where vapor desublimates.

Figure 5 [in draft] placed, with caption.

2. **Computing the conductivity** (L361–370). [cut] L363–367 repeats Section 2.3. [incorrect] "off diagonal components are at least one order of magnitude lower" (L368–370): k_{xy} reaches about 12% of k_{xx} on one packing; it is zero in the mean. [missing] Anisotropy: state k_{xx}/k_{yy}, including 0.79 and 0.64 at the two highest porosities.

3. **Temperature sets the clock** (L371–379). [in draft] panels (b)–(d) in words. [missing] Panel (a) is not mentioned. [missing] **The result as a number**: for every packing, k_{eff} at matched SSA agrees across the five temperatures within 0.27% (median 0.07%); the rate follows τ_{sub} within 4%. [missing] How much the conductivity changes: from day 1 to day 30 it rises by 7–12% at $-40\,^{\circ}$C and by 31–44% at $-5\,^{\circ}$C (the table by porosity and temperature). [revise] "the trend is linear" (L379): linear in $\log\theta$, about 0.13 per decade. [missing] Why the curves are normalized at $\theta_r = 30$ and not at $t = 0$: the first day depends on the interface parameter.

Figure 6 [in draft] placed. [revise] Caption says $\lvert\sigma\rvert \leq 0.31$; the bar reads 0.34. The symbol θ_r on panels (c), (d) is not defined in the caption.

4. **Porosity** (L380–398). [in draft] Structure is right: (a), why two porosities are set aside, (b), (c), (d). [incorrect] "3 packings for each porosity" (L382): five packings at each of three porosities. [incorrect] "We show the time evolution of k_{eff} for each of the simulations" (L383): panel (a) is $-20\,^{\circ}$C only, the mean of five packings. [incorrect] "the ice matrix remains disconnected in the horizontal direction" (L385–387): not shown. What is shown: a weaker and more scattered exponent, and k_{xx}/k_{yy} of 0.79 and 0.64. [revise] L390–391 breaks off ("scientifically their connectivity as"). [incorrect] "The linear trend" in (c) (L393): the axis is logarithmic; the fit is exponential in porosity. [missing] Why four temperatures and not five: the $-40\,^{\circ}$C runs end at $\theta = 43$, too soon after the reference age. [missing] The numbers. Exponent -0.80; per porosity -0.79, -0.82, -0.78, then -0.73 and -0.63. Level $k_{\mathrm{eff},r}/k_{\mathrm{ice}} = 1.37\exp(-5.01\,\varphi)$, 6.6% scatter between packings. At 30 d the conductivity differs by a factor of about 3.3 across porosity. [missing] That the dashed lines in (c), (d) are fitted to $\varphi \leq 0.375$.

Figure 7 **[in draft]** placed. **[revise]** Caption for (a): "a single seed" with "seed scatter" (it is the mean of five packings and their range). Caption does not mention the arrow or the gray band.

Left out of the Results that the data support:

- One number that sums up the collapse: at $\theta = 331$, $k_{\mathrm{eff}}/k_{\mathrm{eff},r} = 1.171 \pm 0.015$ and $\mathrm{SSA}/\mathrm{SSA}_r = 0.819 \pm 0.007$ over the packings with $\varphi \leq 0.375$.
- The temperature dependence of the clock as an activation energy, about 50 $\mathrm{kJ\,mol^{-1}}$.
- That the scatter in the exponent does not shrink with domain size while the scatter in the level does (11% to 4.5%), so the first is a property of the packing and the second of sampling.
- That the exponent of a packing follows how efficiently it conducts (correlation -0.77 with $k_{\mathrm{eff},r}$ over the upper bound at its porosity). *Open: whether to show this; it is the only quantitative support for "connectivity explains the fan".*

4 Discussion

State of the draft. Unchanged since the last check. The opening (L400–410) is the third summary of the plan. Section 4.1 (L412–491) argues that $\beta_{\mathrm{sub}}(T)$ follows Libbrecht's $\sigma_0(T)$, that the model was tuned once and that $-5\,^{\circ}\mathrm{C}$ is "a genuine prediction"; with a constant α_c none of this holds. Section 4.2 (L492–626) claims a collapse across porosity, isotropy and sintering "on the order of decades".

Statements to remove wherever they survive.

- **[incorrect]** L428–440: mobility fitted; coefficient "determined a priori" from single-crystal measurements; "genuine prediction".
- **[incorrect]** L448–449: Wilson–Frenkel described as the diffusion-limited law. It is the attachment-limited one.
- **[incorrect]** L456–468: $\sigma_0(T)$ gives β_{sub} its temperature dependence; "successful parameterization".
- **[incorrect]** L525: "our systems remain isotropic".
- **[incorrect]** L532–536: the collapse is expected because SSA is "the only changing parameter". Conductivity depends on the whole geometry; the collapse across temperature holds because one packing follows one geometric path at every temperature.
- **[incorrect]** L541–545: curves at different porosity "collapse towards a master curve"; packings "converge towards similar configurations".
- **[revise]** L538–539: an author note left in the text.
- **[incorrect]** L582: "on the order of decades". L599: conductivity "will start near 0.0". L623: "the temperatures for which our model was calibrated".

Suggested construction, about twelve paragraphs. Each says what it argues and what in the draft can be reused.

4.1 What vapor transport alone reproduces

1. **The unsaturated pair.** With α_c at the top of its range, the vapor route gives 41% of the measured growth at $-20\,^{\circ}\mathrm{C}$; the rest needs another route. At $-5\,^{\circ}\mathrm{C}$ it gives more than was measured, and the authors doubt that data set. The two cannot both be closed by one α_c.
2. **The saturated pair.** The exponent, against Demmenie's and against 1/3. *Open: which reading, as in Section 3.1.*
3. **α_c is an effective parameter.** Reuse L469–491, the survey of Bouvet and Braun, shortened by half. What it means for the aggregates: their rates carry the uncertainty of α_c; the conductivity as a function of SSA does not.

4.2 Temperature as a rescaling of time

4. **Why the collapse holds.** One route, a constant α_c, transfer limited by attachment: temperature enters through τ_{sub} alone. In this model it is close to a consequence of the construction; say so.
5. **What would break it.** A temperature-dependent α_c; a second route with a different activation energy; a temperature gradient. The clock's activation energy, about 50 $\mathrm{kJ\,mol^{-1}}$, against 24.3 ± 3.3 from Choukroun et al.: the difference is itself evidence of another route in their samples.

4.3 Specific surface area as the state variable

6. **Density is not enough.** Reuse L498–517 (Calonne 2011, 2014), cut to half: conductivity changes at constant density there and here.
7. **The power law and its limit.** One exponent for $\varphi \leq 0.375$; porosity sets the level. Above that the exponent weakens and scatters, and the packings are anisotropic.
8. **Why packings fan out.** Connectivity, not sampling: the scatter in the exponent does not fall with domain size. The two-dimensional point: discs at these porosities are close to the limit of a connected packing. *Open: the percolation and packing thresholds for this argument are not yet checked against a source.*

4.4 What a laboratory could measure

9. Three tests on unloaded ice-sphere samples: superposition of curves at different temperatures and its activation energy; k_{eff} against SSA; SSA against log time. Reuse L550–556 on how SSA is measured.

4.5 Implications for plume deposits on Enceladus

10. Qualitative only. Reuse L568–590 without "decades" and L601–617 (Choukroun), cut by a third: as deposits sinter, SSA falls and conductivity rises together, and SSA is the quantity to track.

11. Reuse L591–600 (radar attenuation) without the starting value. State the two gaps to the moon: temperature (the runs stop at $-40\,^{\circ}\mathrm{C}$) and grain size (50 μm against micron-sized). No extrapolation.

4.6 Limitations

12. Two dimensions; air-filled sealed pores at 1 atm; one grain size; the vapor route only; the first day depends on the interface parameter; the power law holds for $\varphi \leq 0.375$. Reuse L557–567: conduction only; constant k_{ice}, k_{air}.

5 Conclusions

- [revise] L628–643 claims "the first pore-scale numerical model that is able to replicate the temperature dependent kinetics" and a single tuning; L647 refers to Libbrecht's data range, which the model no longer uses.
- Two paragraphs: the three results, each with its range of validity; then the experiment and the model extension that would test them (a second transport route; three dimensions).

Open Research; Acknowledgments

[missing] Code and data availability; run inventory (`RUN_TABLE.md`). Acknowledgments heading is empty (L658).

Supporting Information

- **Tables.** S1 complete simulation parameters (if moved out of the main text); S2 two-grain parameters, with the saturated pair.
- **Figures.** S1 domain-size convergence; S2 interface parameter; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Text.** Packing construction; the weak-form derivation, if cut from Section 2.3; run table.